

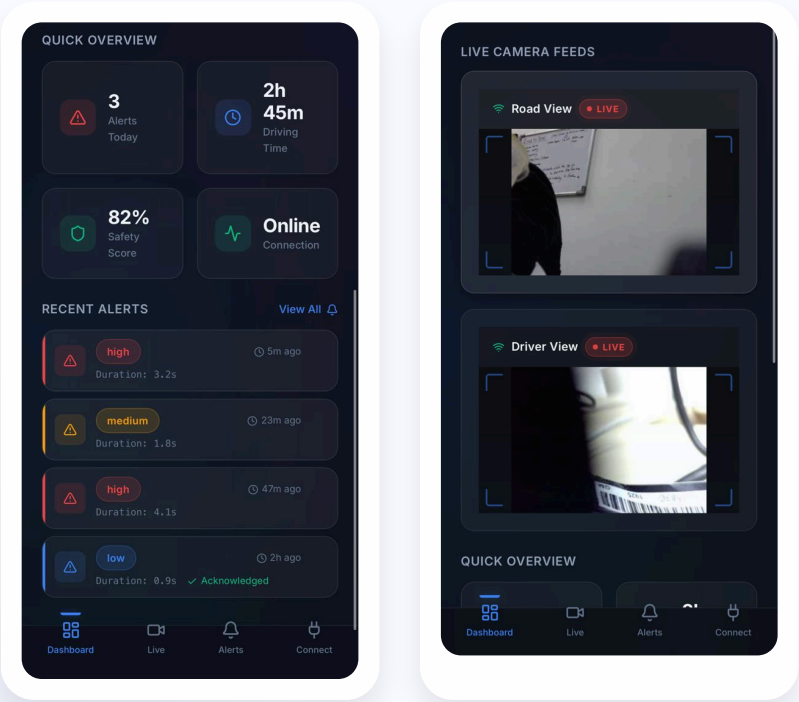
Design Requirements

REQUIRED DESIGN FEATURES AND BASIC FUNCTIONALITIES

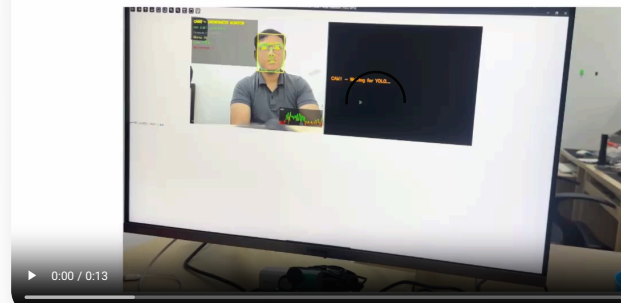
Based on the validated user needs from professional driver interviews and root cause analysis, DashGUARD requires the following core design features and functionalities to effectively address the microsleep crisis on Malaysian roads:

CORE SAFETY AND FUNCTIONALITY	FUNCTIONALITY	USER NEED ADDRESSED	JUSTIFICATION
1 Real-Time Physiological Fatigue Detection	Continuous monitoring of eye closure ratio (PERCLOS), blink frequency, and head position using camera	Drivers need warning BEFORE microsleep strikes at highway speeds	RapidKL driver said accident reduction potential with reliable alerts; research validates PERCLOS as gold standard for drowsiness detection
2 Progressive 3-Stage Alert System	Escalating notifications (Stage 1: LED pulse → Stage 2: Audible alert → Stage 3: App notification)	Wake driver without panic; avoid startle effect	Both drivers emphasized need for "gentle, non-startling alerts that maintain awareness rather than panic"
3 Plug-and-Play Installation	USB or lighter plug power; universal rear camera mount; no wiring required	Easy installation for all vehicles	Grab driver emphasized zero installation costs critical for adoption

App Features



Tested Prototype



Minimum Viable Product

